How-To Library

Informatica

Address Verification Best Practices for Brazil

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Abstract

This document tells you how to connect the elements in a Brazil address to the appropriate address elements in Informatica® Address Verification. This document also describes best practices that you can implement when you verify addresses in Brazil.

Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.x

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Introduction

Informatica Address Verification can validate and enhance addresses in Brazil. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Brazil address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

You must map each item of address information to an appropriate element in Address Verification.

Process Modes

Informatica Address Verification supports the following process modes for Brazil addresses:

- BATCH
- COUNTRY RECOGNITION
- FAST COMPLETION
- INTERACTIVE

- PARSE

Language and Character Set

Brazil reference address databases store address data in the Latin-1 script. By default, Informatica Address Verification returns the verified Brazil addresses in the Latin-1 script.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The Brazil postal service recognizes the following structure for a postal address:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: STREET HOUSE NUMBER
Line 3: CITY [-] STATE CODE
Line 4: POSTAL CODE
Line 5: COUNTRY
```

The following sample address illustrates the structure:

```
BOULEVARD DAS FLORES, 255
SALVADOR-BA
40301-110
BRA
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- Locality
- Postal Code
- Street

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

Information Type	Address Information	Address Elements
Street Name	BOULEVARD DAS FLORES	Street
House Number	255	HouseNumber
Postal Code	40301-110	PostalCode
City	SALVADOR	Locality
State Code	BA	ProvinceAbbreviation

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the address information to the XML address elements in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
    <AddressElements>
        <Country Item="1" Type="NAME">BRAZIL</Country>
        <Locality Item="1" Type="COMPLETE">SALVADOR</Locality>
        <PostalCode Item="1" Type="UNFORMATTED">40301-110</PostalCode>
        <Province Item="1" Type="COUNTRY_STANDARD">BA</Province>
        <Street Item="1" Type="COMPLETE">BOULEVARD DAS FLORES</Street>
        <Number Item="1" Type="COMPLETE">255</Number>
    </AddressElements>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.13.1.45876" -->

<Result     ProcessStatus="V2"
     ModeUsed="BATCH"
     Count="1"
     CountOverflow="NO"
     CountryISO3="BRA"
     PreferredScript="LATIN_1"
     PreferredLanguage="ENGLISH">

<ResultData ResultNumber="1"
     MailabilityScore="4"
     ResultPercentage="100.00"
     ElementResultStatus="F0F8F0F04000000000E0"
     ElementInputStatus="60606060200000000060"
     AddressResolutionCode="00000000000000000000"
     ExtElementStatus="00000000B00000000000"
     ElementRelevance="10101010100000000010"
     AddressType="S"
     LanguageISO3="POR">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">Brazil</Country>
        <Locality Item="1">Salvador</Locality>
        <Locality Item="2">Barbalho</Locality>
        <PostalCode Item="1">40301-110</PostalCode>
        <Province Item="1">BA</Province>
        <Street Item="1">Boulevard das Flores</Street>
        <Number Item="1">255</Number>
    </AddressElements>
    <AddressLines>
        <DeliveryAddressLine Line="1">Boulevard das Flores, 255</DeliveryAddressLine>
        <CountrySpecificLocalityLine Line="1">Salvador-BA</CountrySpecificLocalityLine>
        <FormattedAddressLine Line="1">Boulevard das Flores, 255</FormattedAddressLine>
        <FormattedAddressLine Line="2">Barbalho</FormattedAddressLine>
        <FormattedAddressLine Line="3">Salvador-BA</FormattedAddressLine>
        <FormattedAddressLine Line="4">40301-110</FormattedAddressLine>
        <FormattedAddressLine Line="5">Brazil</FormattedAddressLine>
    </AddressLines>
    <AddressComplete>Boulevard das Flores, 255
Barbalho
Salvador-BA
40301-110
Brazil</AddressComplete>
</ResultData>
</Result>
```

Input and Result XML in Address Verification

The following example shows how you can verify a Brazil address in a SOAP call to Address Verification.

You submit the following request:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:v="http://
validator5.AddressDoctor.com/Webservice5/v4">
 <soap:Header/>
 <soap:Body>
    <v:Process>
        <v:login>****</v:login>
        <v:password>****</v:password>
        <v:parameters>
            <v:ProcessMode>BATCH</v:ProcessMode>
        <v:ServiceParameters></v:ServiceParameters>
        <v:ValidationParameters>
        </v:ValidationParameters>
        </v:parameters>
        <v:addresses>
          <v:Address>
               <v:FormattedAddress>
                    <v:string>BOULEVARD DAS FLORES, 255</v:string>
                    <v:string>SALVADOR-BA</v:string>
                    <v:string>40301-110</v:string>
                    <v:string>BRAZIL</v:string>
               </v:FormattedAddress>
          </v:Address>
        </v:addresses>
        <v:enrichments>
        </v:enrichments>
    </v:Process>
 </soap:Body>
</soap:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://www.w3.org/2003/05/soap-envelope" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
   <soap:Body>
      <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
         <ProcessResult>
            <StatusCode>100</StatusCode>
            <StatusMessage>OK</StatusMessage>
            <Results>
               <Result>
                  <ProcessStatus>V2</ProcessStatus>
                  <CountryISO3>BRA</CountryISO3>
                  <ResultDataSet>
                     <ResultData>
                        <ResultNumber>1</ResultNumber>
                        <MailabilityScore>4</MailabilityScore>
                        <ResultPercentage>100.00</ResultPercentage>
                        <ElementInputStatus>6060606020000000060</ElementInputStatus>
                        <ElementResultStatus>F0F8F0F04000000000E0</ElementResultStatus>
                        <ElementRelevance>1010101010000000010</ElementRelevance>
                        <ExtElementStatus>00000000B00000000000</ExtElementStatus>
                        <AddressResolutionCode>00000000000000000000</AddressResolutionCode>
                        <AddressType>S</AddressType>
                        <LanguageISO3>POR</LanguageISO3>
                        <Address>
                           <Street>
                              <string>BOULEVARD DAS FLORES</string>
                           </Street>
                           <HouseNumber>
                              <string>255</string>
                           </HouseNumber>
                           <Locality>
                              <string>SALVADOR</string>
                              <string>BARBALHO</string>
                           </Locality>
```

```
                        <PostalCode>
                           <string>40301-110</string>
                        </PostalCode>
                        <Province>
                           <string>BA</string>
                        </Province>
                        <Country>
                           <string>BRAZIL</string>
                        </Country>
                        <DeliveryAddressLines>
                           <string>BOULEVARD DAS FLORES, 255</string>
                        </DeliveryAddressLines>
                        <CountrySpecificLocalityLine>
                           <string>SALVADOR-BA</string>
                        </CountrySpecificLocalityLine>
                        <FormattedAddress>
                           <string>BOULEVARD DAS FLORES, 255</string>
                           <string>BARBALHO</string>
                           <string>SALVADOR-BA</string>
                           <string>40301-110</string>
                           <string>BRAZIL</string>
                        </FormattedAddress>
                        <AddressComplete>BOULEVARD DAS FLORES, 255;BARBALHO;SALVADOR-BA;
40301-110;BRAZIL</AddressComplete>
                     </Address>
                  </ResultData>
               </ResultDataSet>
            </Result>
         </Results>
      </ProcessResult>
   </ProcessResponse>
</soap:Body>
</soap:Envelope>
```

Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Brazil addresses.

The following table lists the types of enrichment that you can add to a Brazil address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid Brazil addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Country-Specific Address Enrichments	You can retrieve Brazilian Institute of Geography and Statistics (IBGE) code as an enrichment for a valid Brazil address. The IBGE code is a seven-digit numeric code that identifies cities and states in Brazil. The IBGE code is useful for e-commerce operations as you can use this code for taxation and audit purposes.
Standard Geocoding	You can retrieve standard, interpolated geocoordinates for valid Brazil addresses.

Reference Address Databases

Informatica updates the Brazil reference address databases monthly. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Brazil addresses:

- `BRA5BI.MD`. Batch and interactive modes.
- `BRA5CA.MD`. CAMEO data.
- `BRA5E1.MD`. Supplementary data.
- `BRA5FC.MD`. Fast completion mode.
- `BRA5GC.MD`. Standard geocoding.

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification runs on Informatica Cloud®.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

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